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|---|---|
| Program : **Diploma in Electronics and Computer Engineering** | |
| Course Code : **4509** | Course Title: **Fundamentals of Electronic Communication lab** |
| Semester: **4** | Credits: **1.5** |
| Course Category: **Program Core** | |
| Periods per week: **3 (L:0 T:0 P:3)** | Periods per semester: **45** |

**Course Objectives:**

- To give hands on experience and learn various electronic components, analogue modulation circuits, pulse modulation and digital modulation circuits.
- To demonstrate the working of various stages of AM and FM and solve the problems associated with it.
- To give hands on experience to implement digital modulation techniques.

**Course Prerequisites:**

| Topic | Course code | Course name | Semester |
|---|---|---|---|
| Testing of active and passive components, familiarization of equipment | 2031 | Fundamentals of Electrical and Electronics Engineering lab | 2 |
| Operation of transistor, diode | 2041 | Basic Electronics | 2 |

**Course Outcomes:**

On completion of the course, the student will be able to:

| COn | Description | Duration (Hours) | Cognitive level |
|---|---|---|---|
| CO1 | Illustrate the working of electronic components in communication system | 3 | Understanding |
| CO2 | Experiment with analog modulation schemes | 12 | Applying |
| CO3 | Develop Pulse modulation circuits | 9 | Applying |

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|---|---|---|---|
| CO4 | Model various digital modulation techniques | 15 | Applying |
| | Lab Exam | 6 | |

**CO – PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 2 | | | | | | |
| **CO2** | 3 | | | | | | |
| **CO3** | 3 | | | | | | |
| **CO4** | 3 | | 2 | 3 | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| **CO1** | **Illustrate the working of electronic components in communication system** | | |
| M1.01 | Familiarize 555, 565, Counter IC CD4016. | 3 | Understanding |
| **CO2** | **Experiment with analog modulation schemes.** | | |
| M2.01 | Construct an AM generator using transistor and measure the modulation index | 3 | Applying |
| M2.02 | Construct AM demodulator using diode and observe the input-output waveforms | 3 | Applying |
| M2.03 | Build an FM generator using 555 IC and observe the input-output waveforms. | 3 | Applying |
| M2.04 | Build an FM generator and demodulator using 565 IC and observe the input-output waveforms | 3 | Applying |
| | Lab Exam – I | 3 | |
| **CO3** | **Develop Pulse modulation circuits** | | |
| M3.01 | Develop a PAM modulator and demodulator circuit. Observe the input-output waveforms | 3 | Applying |
| M3.02 | Develop a PWM modulator using 555 IC and observe the input-output waveforms | 3 | Applying |

| | | | |
|---|---|---|---|
| M3.03 | Develop a PPM modulator circuit using 555 and observe the input-output waveforms | 3 | Applying |
| **CO4** | **Model various digital modulation techniques.** | | |
| M4.01 | 1. Construct a simple sampling circuit and observe the waveforms.<br>2. Reconstruct a sine wave from its samples | 3 | Applying |
| M4.02 | Model a Binary Amplitude Shift Keying (modulator & demodulator) and observe the input-output waveforms | 3 | Applying |
| M4.03 | Model a Binary Frequency Shift Keying (modulator & demodulator) and observe the input-output waveforms | 3 | Applying |
| M4.03 | Model a Binary Phase Shift Keying (modulator & demodulator) and observe the input-output waveforms | 3 | Applying |
| M4.04 | Open ended experiment. | 3 | Applying |
| | Lab Exam – II | 3 | |

**** - Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 3-5. There should be no duplication in experiments between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on open ended experiment of their choice.)

Example:

1. Setup an FM receiver in the range 80MHz-108MHz
2. Simulate the Analog modulation scheme.
3. Simulate the digital modulation scheme.

**Text / Reference:**

| T/R | Book Title/Author |
|---|---|
| T1 | Electronic Communication Systems – George Kennedy – TMH |
| T2 | Electronics Lab Manual Vol-2 KA Navas 4th edition |
| R1 | Electronic communications - Roddy and Coolen – PHI |
| R2 | Principles of Communication systems – Taub and Schilling |

**Online Resources:**

| Sl.No | Website Link |
|---|---|
| 1 | https://nptel.ac.in/courses/117/105/117105143 |
| 2 | https://nptel.ac.in/courses/117101051/ |